

2. Requirement Analysis Phase

ComicCraft – AI Comic Story Creator

Introduction

The requirement analysis phase converts the ComicCraft idea into specific functional, technical, frontend, security, and documentation requirements.

Functional Requirements

The system shall accept a story prompt, character name or description, setting, tone, art style, panel count, and additional instructions. It shall generate a structured panel-by-panel story containing narration, dialogue, scene descriptions, and image prompts. It shall generate comic illustrations, display the completed comic, regenerate a selected panel, and provide a downloadable PDF.

Technical Requirements

The specified technology stack consists of Python 3.11+, FastAPI with Uvicorn, Google Gemini API for story generation, Hugging Face Diffusers/Stable Diffusion for images, Pillow for image processing, ReportLab for PDF creation, HTML/CSS/JavaScript for the frontend, and python-dotenv for environment configuration.

API Requirements

The application requires GET / for the website, GET /api/health for health checking, POST /api/generate-story for Gemini story generation, POST /api/create-comic for complete comic creation, POST /api/regenerate-panel for regenerating one panel, and GET /api/comic/{comic_id} for retrieving comic information.

Security Requirements

API keys must be stored only in .env and never hard-coded. Inputs and panel count must be validated, with the panel count restricted to 1–12. Safe filenames and file paths must be used, and API keys, stack traces, and arbitrary local files must not be exposed.

Documentation Requirements

The project must include requirements.txt, .env.example, .gitignore, run.py, and README.md. The README must cover installation, virtual-environment setup, API-key configuration, application execution, image-generation requirements, and troubleshooting.

Project reference: ComicCraft – AI Comic Story Creator